A note on AI: We are aware that all of these tasks can be solved by AI within seconds. However, we encourage you to work on the tasks on your own first and, whenever you get stuck, ask us (or AI) for help. If you just let AI write the solutions, you won't deeply understand what you are doing – and we will notice this directly when we ask you to present your work.

1 Load Data Sets

1. Load the Handball, UEFA Euro Football, and Tennis data into your current R session.
2. Inspect the data and try to get some information from it, e.g., the average number of goals across all matches. Play around with the data. In particular:
   - Get an overview of the structure of the data sets (number of observations, variables and their types), e.g., using `str()` and `summary()`.
   - Check whether there are missing values.
   - Compute some descriptive statistics, e.g., the average number of goals per team and match in the Euro data, the average goal difference in the handball data, or the share of won matches in the tennis data.
   - Visualize the response variables (`scorediff`, `Goals`, and `win`) as well as their relation to selected predictors, e.g., using histograms, bar plots, boxplots or scatter plots. Which predictors seem to be related to the response?

2 Linear Regression

1. Fit the linear regression model from the theory slides (available in Moodle; predictors `bet.probs.diff` and `season`) to the handball data (`handball_wide`).
2. Check the assumptions of the linear regression model, e.g., by applying the R function `plot()` to the fitted model. Are the assumptions reasonably fulfilled?
3. Use the model to predict the goal difference for each match (R function `predict()`). Compare the predicted goal difference to the real values from the data set.
   - Compute the absolute differences between the true and the predicted values. What is the mean and the median?
   - Create a boxplot for the absolute difference between true and predicted values.
4. Take a closer look at the boxplot (and the residual plots from above). What do you notice? Deduce reasonable strategies to change your model accordingly. Fit an improved model and compare the mean and the median of the absolute differences to the ones from your first model.

3 Poisson Regression

1. Fit a Poisson model with all predictors to the Euro_train data.
2. Check the assumptions of the Poisson model. In particular, compare the mean and the variance of the number of goals (equidispersion).
3. Predict the number of goals per match in the Euro_train data and compute the average absolute difference between the predicted and true number of goals.

4. Use the fitted model to predict the number of goals in the Euro_test data (using the `newdata` argument in the `predict()`-function). Again, compute the average absolute difference between the predicted and true number of goals.

5. Is the result an expected or an unexpected result?

6. Try to improve your model. Which steps could lead to an improvement? After you fitted a new (hopefully improved) model, check this by again comparing the difference between prediction and true number of goals in the **test data**.

4 Logistic Regression

1. Load the Tennis data and split them into a training and a test data set, where the data from 2011 to 2017 serve as training data and the data from 2018 to 2024 serve as test data.

2. Fit a Logistic regression model to the training data, using the predictors `rank`, `points`, `elo`, and `age.30`. Interpret the estimated coefficients. Which predictors seem to be relevant?

3. Use the fitted model to predict the matches from the test data set. Compute the ratio of correctly predicted matches!

4. Why do we split the data by time instead of randomly into training and test data? What could be problematic about a random split here?